CARBAMAZEPINE

HLA-B*1502 — SJS

Strong enzyme inducer

CHEMICAL NAME 5H-dibenzo[b,f]azepine-5-carboxamide
CLASS Iminostilbene antiepileptic and mood stabiliser
TRADE NAMES Tegretol · Tegretol Retard · Carbatrol · Epitol

THERAPEUTIC USES / INDICATIONS

- Focal seizures (± secondary generalisation); generalised tonic-clonic seizures.
- Trigeminal neuralgia (first-line).
- Bipolar disorder (prophylaxis, second-line).
- Diabetic neuropathic pain; alcohol withdrawal (adjunct).

ADMINISTRATION

- **PO** with or after food; MR tablets swallowed whole (may halve).
- Start low and increase slowly (autoinduction).
- Prescribe by brand; liquid absorbed faster (more side effects).

DOSAGES

- **Epilepsy:** start 100–200 mg once or twice daily; increase by 100–200 mg every 2 weeks; usual 0.8–1.2 g/day (max 2 g) in divided doses.
- **Trigeminal neuralgia:** 100 mg 1–2 times daily, usual 200 mg 3–4 times daily.
- **Children:** 5 mg/kg/day initially, maintenance 10–20 mg/kg/day.
- **Target level:** 4–12 mg/L.

PHARMACOKINETICS

- **Peak:** 4–8 h (slow, erratic).
- Metabolised by CYP3A4 to active 10,11-epoxide; induces its own metabolism (t½ falls over 2–4 weeks).
- Excreted in urine. **t½:** 25–65 h initially → 12–17 h after autoinduction.

PHARMACODYNAMICS

- Blocks **voltage-gated sodium channels** → stabilises hyperexcitable neuronal membranes.
- Anticholinergic and antidiuretic (ADH-like) effects.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dizziness, drowsiness, ataxia, diplopia (dose-related).
- Nausea, vomiting.
- Rash (common).

ADVERSE EFFECTS (SERIOUS)

- SJS/TEN; HLA-B*1502 (Han Chinese, Thai, other Asian); DRESS (HLA-A*3101).
- Hyponatraemia (SIADH).
- Aplastic anaemia, agranulocytosis, leucopenia; hepatitis.
- Arrhythmias, AV block; teratogenic (neural tube defects).

CONTRAINDICATIONS & CAUTIONS

- AV conduction abnormalities (unless paced).
- History of bone-marrow depression; acute porphyria.
- Use with MAOIs (within 14 days).
- HLA-B*1502 positive (avoid).

KEY INTERACTIONS

- Enzyme inducer: ↓ oral contraceptives, warfarin, DOACs, steroids, antipsychotics, antidepressants, other antiepileptics, immunosuppressants.
- ↑ **carbamazepine:** erythromycin, clarithromycin, fluconazole, isoniazid, verapamil, diltiazem, grapefruit.
- **Lamotrigine, valproate:** complex interactions.

NURSING MANAGEMENT

- Check HLA-B*1502 in at-risk ancestry before starting.
- Baseline and periodic CBC, LFTs, Na^+, renal function, levels.
- Monitor for rash, fever, sore throat, bruising.
- Seizure diary; dose titration.
- Watch for hyponatraemia (confusion, headache) in elderly.

PATIENT EDUCATION

- Report rash, fever, mouth ulcers, bruising, sore throat immediately.
- Hormonal contraception may fail — use IUD/injection; folic acid before pregnancy.
- Avoid alcohol and grapefruit; take with food.
- Do not stop suddenly; may cause drowsiness/double vision.

KEY POINTS

- Na^+-channel blocker; trigeminal neuralgia first-line.
- SJS (HLA-B*1502), hyponatraemia, marrow suppression.
- Autoinduction; titrate slowly.
- Strong enzyme inducer — pill failure.